Lakshya JEE 2.0 2025

Maths

Determinants

DPP: 1

Q1 The value of the determinant $\begin{vmatrix} 1 & \log_b a \\ \log_a b & 1 \end{vmatrix}$ is equal to:

(A) 1

(B) $\log_a b$

(C) $\log_b a$

(D) 0

Q2 If ω is non-real complex cube root of unity, then $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & \omega & 1 \end{vmatrix}$ is equal to

(A) 0

(B) 1

(C) 3

(D) 2

Q3 If $\mathrm{A}+\mathrm{B}+\mathrm{C}=\pi$, then value of $\begin{vmatrix} \sin(A+B+C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$ is

(A) 0

(B) 1

(C) $2\sin B\cdot\tan A\cos C$

(D) $2\sin A\sin B\cdot\sin C$

Q4 The equation $\begin{vmatrix} x-2 & 3 & 1 \\ 4x-2 & 10 & 4 \\ 2x-1 & 5 & 1 \end{vmatrix}=0$ is satisfied by

(A) $x=-2$

(B) $x=-5$

(C) $x=-7$

(D) $x=-9$

Q5 The value of the determinant $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$ is

(A) 0

(B) 1

(C) 2

(D) 3

Q6 The value of the determinant $\begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix}$ is equal to

(A) $(x-y)(y-z)(z-x)$

(B) $(x-y)(y-z)(z-x)(x+y+z)$

(C) $(x+y+z)$

(D) $(x-y)(y-z)(z-x)(xy+yz+zx)$

Q7 The value of $\begin{vmatrix} 5^2 & 5^3 & 5^4 \\ 5^3 & 5^4 & 5^5 \\ 5^4 & 5^5 & 5^6 \end{vmatrix}$, is

(A) 5^2

(B) 0

(C) 5^{13}

(D) 5^9

Q8 The value of $\begin{vmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \\ 4 & 4 & 3 & 0 & 0 \\ 5 & 5 & 5 & 4 & 0 \\ 6 & 6 & 6 & 6 & 5 \end{vmatrix}$ is

(A) 6!

(B) 5!

(C) $1.2^2\cdot 3\cdot 4^3\cdot 5^4\cdot 6^4$

(D) None of these

Q9 The cofactor of the element 4 in the determinant $\begin{vmatrix} 1 & 3 & 5 & 1 \\ 2 & 3 & 4 & 2 \\ 8 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \end{vmatrix}$ is

(A) 4 (B) 10
(C) -10 (D) -4

Q10 If $s=(a+b+c)$, then value of $\begin{vmatrix} s+c & a & b \\ c & s+a & b \\ c & a & s+b \end{vmatrix}$ is

(A) $2s^2$ (B) $2s^3$
(C) s^3 (D) $3s^3$

Q11 If a, b, c are positive and not equal then value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ may be:

(A) 1 (B) -3
(C) 2 (D) 4

Q12 The parameter, on which the value of the determinant $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ does not depend upon, is:

(A) a (B) p
(C) d (D) x

Q13 $\begin{vmatrix} (a^x+a^{-x})^2 & (a^x-a^{-x})^2 & 1 \\ (b^x+b^{-x})^2 & (b^x-b^{-x})^2 & 1 \\ (c^x+c^{-x})^2 & (c^x-c^{-x})^2 & 1 \end{vmatrix}$ is equal to:

(A) 0
(B) $2abc$
(C) $a^2b^2c^2$
(D) abc

Q14

The roots of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix}=0$ are:

(A) $-1,-2$ (B) $-1,2$
(C) $1,-2$ (D) $1,2$

Q15 If $a>0, b>0, c>0$ are respectively the $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ terms of a G.P., then the value of the determinant $\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix}$ is

(A) 1
(B) -1
(C) $\frac{abc}{pqr}$
(D) 0

Q16 The value of the determinant $\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(a+c) \\ 1 & ab & c(a+b) \end{vmatrix}$ doesn't depend on

(A) a (B) b
(C) c (D) $a+b+c$

Q17 Let $P=[a_{ij}]$ be a 3×3 matrix and let $Q=[b_{ij}]$, where $b_{ij}=2^{i+j}a_{ij}$ for $1<\mathrm{i}\,,\ \mathrm{j}\leq \mathrm{i},\mathrm{j}\leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is

(A) 2^{10} (B) 2^{11}
(C) 2^{12} (D) 2^{13}

Q18 The value of θ satisfying the equation $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix}=0$ are:

(A) $\frac{4\pi}{24}, \frac{3\pi}{24}$
(B) $\frac{7\pi}{24}, \frac{5\pi}{24}$
(C) $\frac{7\pi}{24}, \frac{11\pi}{24}$
(D) $\frac{\pi}{24}, \frac{11\pi}{24}$

Answer Key

Q1	(D)
Q2	(A)
Q3	(A)
Q4	(C)
Q5	(A)
Q6	(B)
Q7	(B)
Q8	(B)
Q9	(B)
Q10	(B)
Q11	(B)
Q12	(B)
Q13	(A)
Q14	(B)
Q15	(D)
Q16	(A, B, C, D)
Q17	(D)
Q18	(C)